

LABSHEET - 4**HIMANI CHAUDHARY****BTECH(CSE) 3RD YEAR****CU240250967****SECTION - B****ROLL NO. - 24****Mastery in Data Analytics and Visualizations and Career Advancement (Technical)**

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
# EXPERIMENT NO. 4
# FREQUENCY DOMAIN IMAGE FILTERING USING FOURIER TRANSFORM
# =====

print("=" * 65)
print("FREQUENCY DOMAIN IMAGE FILTERING USING FOURIER TRANSFORM")
print("=" * 65)
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
FREQUENCY DOMAIN IMAGE FILTERING USING FOURIER TRANSFORM
=====
Himani Chaudhary
CU240250967
```

```
# 1. LOAD IMAGE
# -----

# Use r before the path
image_path = "/content/beautiful-nature-mountain-scenery-with-flowers-picjumbo-com.jpg"
image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
    print("Check your image path.")
    exit()

print("\nOriginal image loaded.")
print("Original image shape:", image.shape)

print("Himani Chaudhary")
print("CU240250967")
```

```
Original image loaded.
Original image shape: (3584, 5376, 3)
Himani Chaudhary
CU240250967
```

```
# 2. CONVERT TO GRAYSCALE
# -----

image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("After grayscale conversion:", image.shape)
print("Himani Chaudhary")
print("CU240250967")
6
```

```
After grayscale conversion: (3584, 5376)
Himani Chaudhary
CU240250967
6
```

```
# 3. RESIZE IMAGE
# -----

image = cv2.resize(image, (800, 600))

print("After resizing:", image.shape)
print("Himani Chaudhary")
print("CU240250967")
```

After resizing: (600, 800)
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CU240250967

```
# 4. CONVERT IMAGE TO FLOAT32
# -----

image_float = np.float32(image)

print("Image data type:", image_float.dtype)
print("Himani Chaudhary")
print("CU240250967")
```

Image data type: float32
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CU240250967

```
# 5. COMPUTE FOURIER TRANSFORM
# -----

dft = cv2.dft(
    image_float,
    flags=cv2.DFT_COMPLEX_OUTPUT
)

print("\nDFT calculated successfully.")
print("Himani Chaudhary")
print("CU240250967")
```

DFT calculated successfully.
Himani Chaudhary
CU240250967

```
# 6. SHIFT ZERO FREQUENCY TO CENTER
# -----

dft_shift = np.fft.fftshift(dft)
print("Himani Chaudhary")
print("CU240250967")
```

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CU240250967

```
# 7. MAGNITUDE SPECTRUM
# -----

magnitude_spectrum = cv2.magnitude(
    dft_shift[:, :, 0],
    dft_shift[:, :, 1]
)

magnitude_spectrum = 20 * np.log(
    magnitude_spectrum + 1
)

magnitude_spectrum = cv2.normalize(
    magnitude_spectrum,
    None,
    0,
    255,
    cv2.NORM_MINMAX
)

magnitude_spectrum = np.uint8(magnitude_spectrum)
```

```
print("Frequency spectrum calculated.")
print("Himani Chaudhary")
print("CU240250967")
```

```
Frequency spectrum calculated.
Himani Chaudhary
CU240250967
```

```
# 8. IMAGE CENTER
# -----
```

```
rows, cols = image.shape

crow = rows // 2
ccol = cols // 2

radius = 40
print("Himani Chaudhary")
print("CU240250967")
```

```
Himani Chaudhary
CU240250967
```

```
# 9. LOW-PASS FILTER
# -----
```

```
low_pass_mask = np.zeros(
    (rows, cols, 2),
    dtype=np.float32
)

cv2.circle(
    low_pass_mask,
    (ccol, crow),
    radius,
    (1, 1),
    -1
)

# Apply filter
low_pass_dft = dft_shift * low_pass_mask

# Shift back
low_pass_ishift = np.fft.ifftshift(low_pass_dft)

# Inverse Fourier Transform
low_pass_result = cv2.idft(low_pass_ishift)

# Magnitude
low_pass_result = cv2.magnitude(
    low_pass_result[:, :, 0],
    low_pass_result[:, :, 1]
)

# Normalize
low_pass_result = cv2.normalize(
    low_pass_result,
    None,
    0,
    255,
    cv2.NORM_MINMAX
)

low_pass_result = np.uint8(low_pass_result)

print("Low-pass filter completed.")
print("Himani Chaudhary")
print("CU240250967")
```

```
Low-pass filter completed.
Himani Chaudhary
CU240250967
```

```
# 10. HIGH-PASS FILTER
# -----
```

```
high_pass_mask = np.ones(
    (rows, cols, 2),
    dtype=np.float32
)

cv2.circle(
    high_pass_mask,
    (ccol, crow),
    radius,
    (0, 0),
    -1
)

# Apply filter
high_pass_dft = dft_shift * high_pass_mask

# Shift back
high_pass_ishift = np.fft.ifftshift(high_pass_dft)

# Inverse Fourier Transform
high_pass_result = cv2.idft(high_pass_ishift)

# Magnitude
high_pass_result = cv2.magnitude(
    high_pass_result[:, :, 0],
    high_pass_result[:, :, 1]
)

# Normalize
high_pass_result = cv2.normalize(
    high_pass_result,
    None,
    0,
    255,
    cv2.NORM_MINMAX
)

high_pass_result = np.uint8(high_pass_result)

print("High-pass filter completed.")
print("Himani Chaudhary")
print("CU240250967")
```

High-pass filter completed.
Himani Chaudhary
CU240250967

```
# 11. SAVE OUTPUTS
# -----

cv2.imwrite("original_image.jpg", image)

cv2.imwrite(
    "frequency_spectrum.jpg",
    magnitude_spectrum
)

cv2.imwrite(
    "low_pass_filtered.jpg",
    low_pass_result
)

cv2.imwrite(
    "high_pass_filtered.jpg",
    high_pass_result
)

print("\nAll output images saved successfully.")
print("Himani Chaudhary")
print("CU240250967")
```

All output images saved successfully.
Himani Chaudhary
CU240250967

```
# 12. DISPLAY RESULTS
# -----
```

```
plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(image, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(magnitude_spectrum, cmap="gray")
plt.title("Frequency Spectrum")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(low_pass_result, cmap="gray")
plt.title("Low-Pass Filtered Image")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(high_pass_result, cmap="gray")
plt.title("High-Pass Filtered Image")
plt.axis("off")

plt.tight_layout()

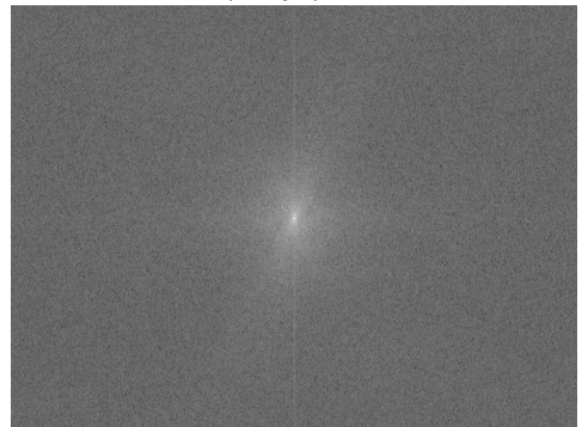
plt.savefig(
    "frequency_filtering_result.jpg",
    dpi=300
)

plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

Original Image



Frequency Spectrum



Low-Pass Filtered Image



High-Pass Filtered Image



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```
# 13. FINAL OUTPUT
```

```
# -----
```

```
print("\n" + "=" * 65)
```

```
print(" EXPERIMENT COMPLETED SUCCESSFULLY ")
print("=" * 65)

print("\nGenerated files:")
print("1. original_image.jpg")
print("2. frequency_spectrum.jpg")
print("3. low_pass_filtered.jpg")
print("4. high_pass_filtered.jpg")
print("5. frequency_filtering_result.jpg")
print("Himani Chaudhary")
print("CU240250967")
```

```
=====
EXPERIMENT COMPLETED SUCCESSFULLY
=====
```

```
Generated files:
1. original_image.jpg
2. frequency_spectrum.jpg
3. low_pass_filtered.jpg
4. high_pass_filtered.jpg
5. frequency_filtering_result.jpg
Himani Chaudhary
CU240250967
```